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AI and machine learning for soil analysis: an assessment of sustainable agricultural practices

M. Awais | Syed Zaigham | +8 authors | Jiandong Hu

Agricultural and Food Sciences, Computer Science · Bioresources and Bioprocessing · 1 December 2023

TLDR The conventional statistical tools for SWCs and soil texture analysis are not enough to practice and manage ergonomic land **management**, so the broader geospatial non-numeric data are more suitable for AI processing that may soon help soil scientists develop a global SWC database. Expand

64 | PDF | Springer Nature | Save | Cite

Domains and Impact of AI in IoT for Environmental Monitoring and Management: A Review

D. S. Hammadi | Dhuha Basheer Abdullah | Environmental Science, Computer Science · International Conference on Computer Science and… · 15 April 2025

This analysis looked through 60 papers between 2019 and 2024 and observed how **artificial intelligence**, particularly its **machine learning** and **deep learning** subfields, reshapes the Internet of Things… Expand

2 | IEEE | Save | Cite

Recent Advances and Emerging Directions in Fire Detection Systems Based on Machine Learning Algorithms

B. Diaconu | Computer Science, Environmental Science · Fire · 17 November 2023

TLDR The objective of this **review** is to present the state of the art in the area of **fire** detection, prevention and propagation modeling with **machine learning** algorithms. Expand

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Deep learning in the construction industry: A review of present status and future innovations

T. Akinosho | Lukumon O. Oyedele | +4 authors | Ashraf A. Ahmed

Engineering, Computer Science · 1 November 2020

TLDR Existing studies that have applied **deep learning** to prevalent construction challenges like structural health monitoring, construction site safety, building occupancy modelling and energy demand prediction are **reviewed**. Expand

450 | PDF | Publisher | Save | Cite

Machine Learning Methods for Small Data Challenges in Molecular Science.

Bozheng Dou | Zailiang Zhu | +7 authors | Guo-Wei Wei | Computer Science, Biology · Chemical Reviews · 29 June 2023

TLDR This **review** summarizes and analyzes several emerging potential solutions to small data challenges in molecular science, including chemical and biological sciences and briefly discusses the latest advances in these methods. Expand

355 | PDF | PubMed | Save | Cite

Survey on Advancing Multimodal Remote Sensing in Precision Agriculture

Manish Shriwas | Khushi Sindhi | Agricultural and Food Sciences, Computer Science · 2024 International Conference on **Artificial**… · 20 December 2024

TLDR The study discovers major trends and possible future paths for the field, including the combination of real-time analytics with AI-driven decision support systems, through a detailed examination of case studies and current research. Expand

1 IEEE Save Cite

Advancements in Hematologic Malignancy Detection: A Comprehensive Survey of Methodologies and Emerging Trends

Rajashree Nambiar Ranjith Bhat B. Achar H V Medicine, Computer Science · TheScientificWorldJournal · 1 January 2025

TLDR A synthesis reveals that DL models consistently outperform ML baselines, achieving classification accuracies above 95% in benchmark datasets, with hybrid models pushing peak accuracy to 99.7%. Expand

1 Wiley Save Cite

The Role of AI in Early Detection of Alzheimer's and Parkinson's Diseases: A Literature Survey

Wejdan H. Alhassun Abdulaziz S. Alothman Sultan A. Alfawaz Medicine, Computer Science · Asian Journal of Research in Computer Science · 4 February 2025

TLDR Key AI methodologies, including **machine learning** (ML), **deep learning** (DL), natural language processing (NLP), and reinforcement **learning** (RL), and their applications in early diagnosis are discussed, highlighting the significance of interdisciplinary efforts and sustained research to unlock AI's full potential in medical applications. Expand

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Plant Care AI an AI Based Solution for Plant Disease Diagnosis and Assistance -A Comprehensive Survey

Abhijit Kulkarni Seema Verma Tanish Shingane Snehal Ghogare Shivkanya Dhupe

Computer Science, Agricultural and Food Sciences · 2024 International Conference on Advances in… · 27 September 2024

TLDR An attempt is made to develop a comprehensive web application tailored for plant care, featuring automated plant disease detection, treatment recommendations based on identified conditions, suggestions for nearby nurseries to acquire necessary materials, and chatbot support for user inquiries. Expand

IEEE Save Cite

A Survey of AI-Enabled Predictive Maintenance for Railway Infrastructure: Models, Data Sources, and Research Challenges

Francisco Javier Bris-Peñalver Randy Verdecia-Peña José I. Alonso

Engineering, Computer Science · Italian National Conference on Sensors · 30 January 2026

TLDR The **survey** highlights the dominant data sources and feature engineering **techniques**, evaluates the model performance across subsystems, and identifies research gaps related to data quality, cross-network generalization, model robustness, and integration with real-time asset **management** platforms. Expand

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